Robotics, Artificial Intelligence
and Real-time Systems

Cognitive Systems: Neural Networks and Applied AI

Part I: Foundations and Theory of Cognitive Systems

Chapter 1
A Brief History of Cognitive Systems

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Topics in this Chapter

From ancient Greece…
… to Silicon Valley

- What is the motivation for building cognitive systems?
- What were the first cognitive systems ever built?
- How is robotics related to cognitive systems?
- What can we expect from cognitive systems in the future?

A Brief History of Cognitive Systems

From Ancient Greece to Silicon Valley

Cognitive Systems

If you don't know what it is, call it a system and if you don't know how it works, call it a process

Cognition

mid-15c., “ability to comprehend,” from Latin cognitionem (nominative cognitio) “a getting to know, acquaintance, knowledge,” noun of action from past participle stem of cognoscere “to get to know, recognize” (see cognizance).

System

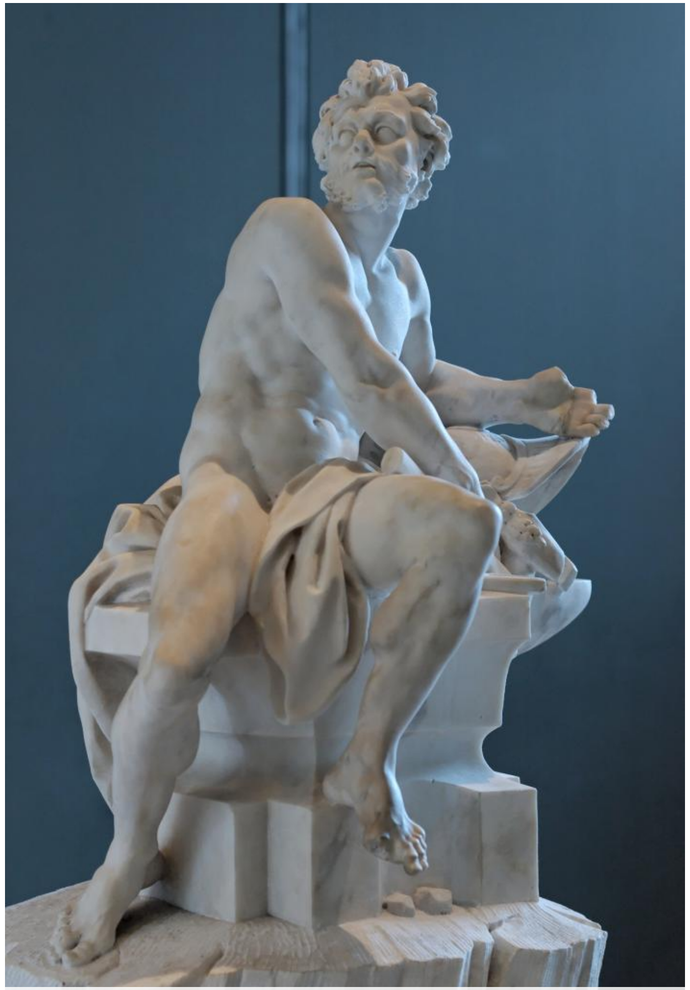
1610s, “the whole creation, the universe,” from Late Latin systema “an arrangement, system,” from Greek systema “organized whole, a whole compounded of parts,” from stem of synistanai “to place together, organize, form in order,” from syn- “together” (see syn-) + root of histanai “cause to stand,” from PIE root *sta- “to stand, make or be firm.”

Online Etymology Dictionary

Machines that Think – An Everlasting Dream

The first descriptions of human-like intelligent machines date back to Greek mythology. It was believed that Hephaestus, the good of fire, blacksmiths, and craftsmen, constructed artificial servants made of gold:

> "And in support of their master moved his attendants. These are golden, and in appearance like living young women. There is intelligence in their hearts, and there is speech in them and strength, and from the immortal gods they have learned how to do things."
>
> *Homer, Iliad 18. 136 ff (trans. Lattimore) (Greek epic C8th B.C.)*
>
> *From http://www.theoi.com/Olympios/HephaistosWorks.html#Automotones*



Marble Statue of Hephaestus by Guillaume Coustou de Younger, 1742. Public Domain

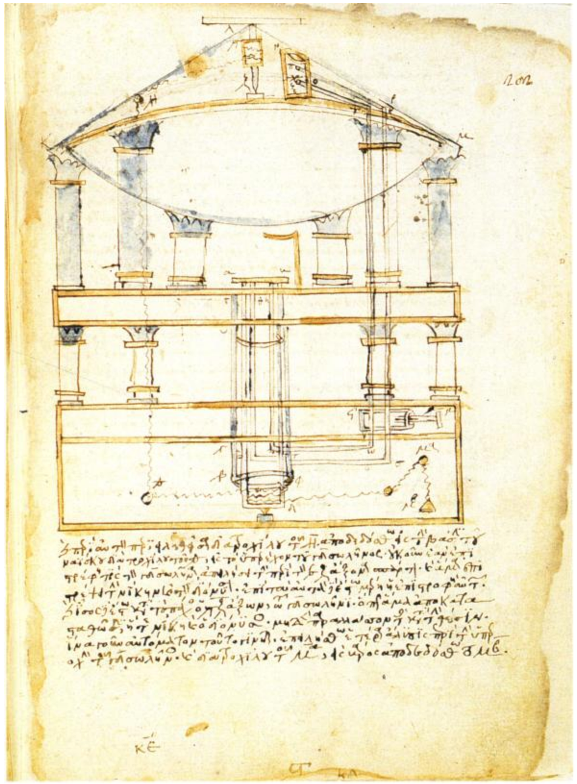
Early Attempts to Build Human-Like Machines

62
Heron of Alexandria builds programmable automata that are used in puppet theaters

1136 – 1206
Ismail al-Jazari develops different humanoid automata that dispense water, flush a basin or play music

1495 – 1497
Leonardo da Vinci draws construction plans for a mechanical knight

1770
Wolfgang von Kempelen constructs the "Mechanical Turk", a human-operated chess-playing fake automaton



Pipes and valves of a wine and water dispensing Bacchus statue in a temple by Heron of Alexandria.

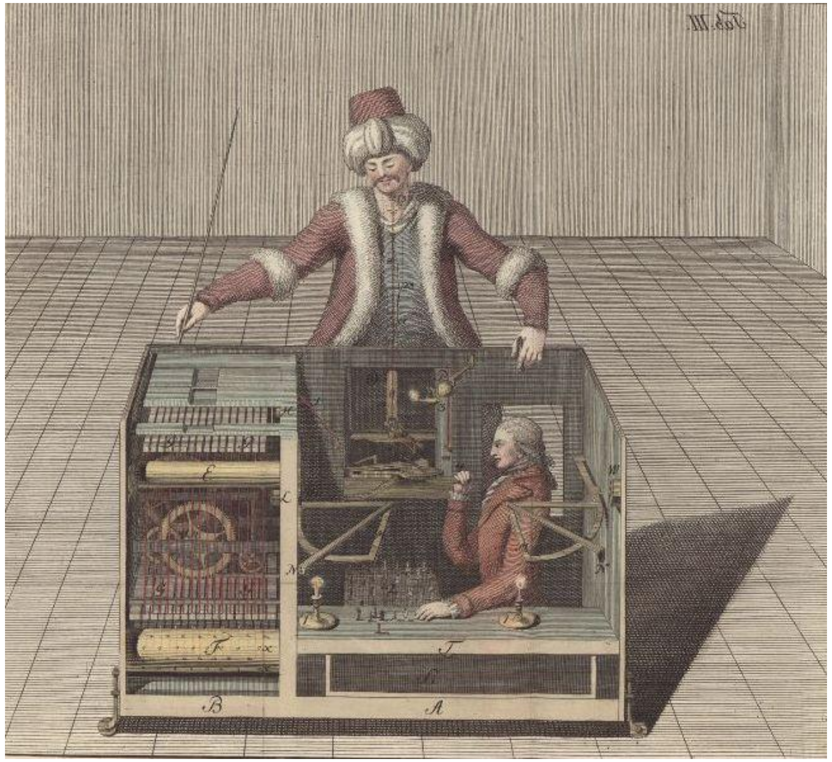
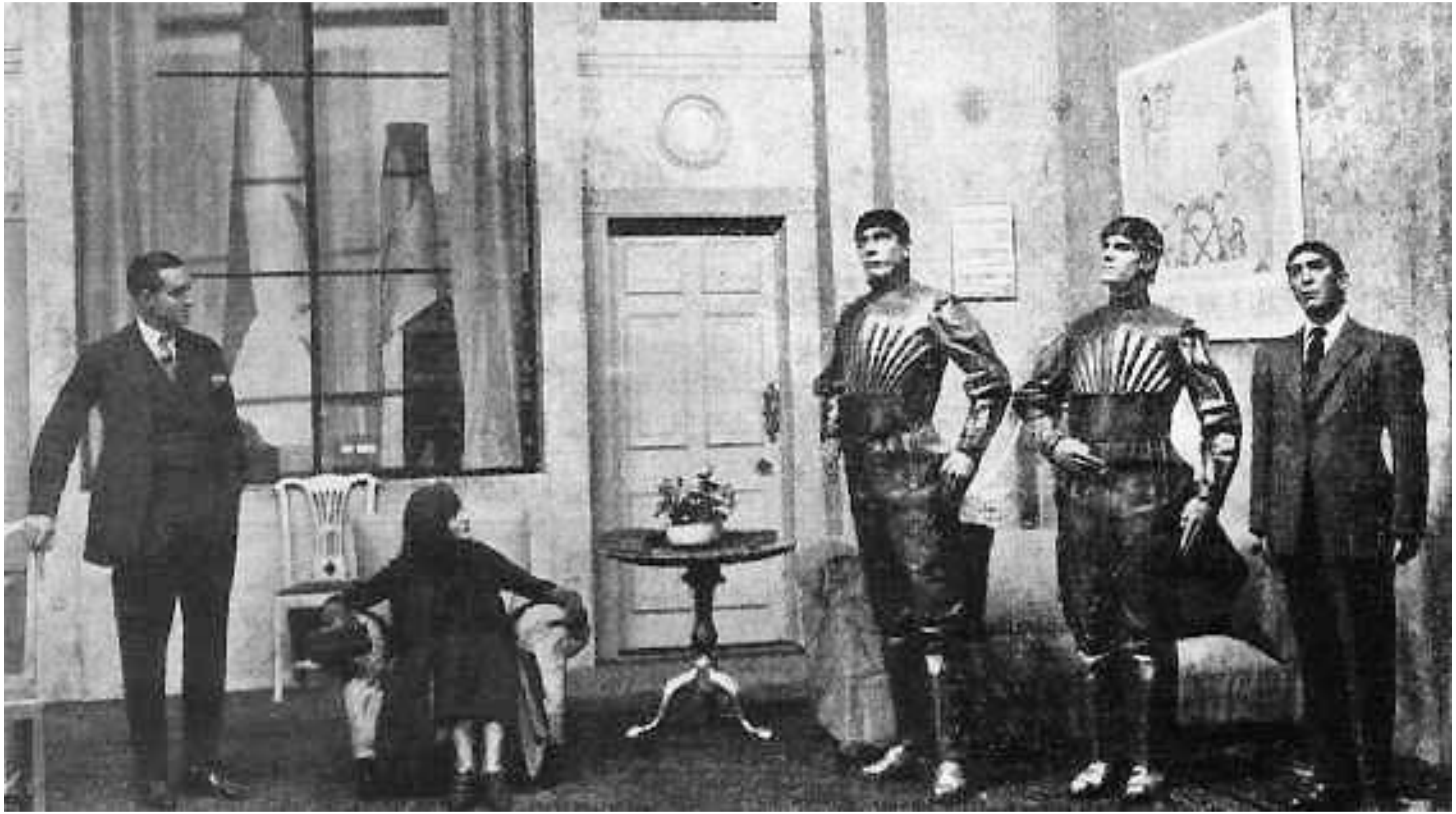


Illustration of a Mechanical Turk

The Birth of Robotics in the 20th Century



1921
The term robot is coined in the play R.U.R (Rossum's Universal Robots) by the Czech writer Karel Čapek.

1942
The three laws of robotics are introduced in Isaac Asimov's short story *Runaround*, which later becomes part of the collection *I, Robot*.

The Beginnings of Humanoid Robotics

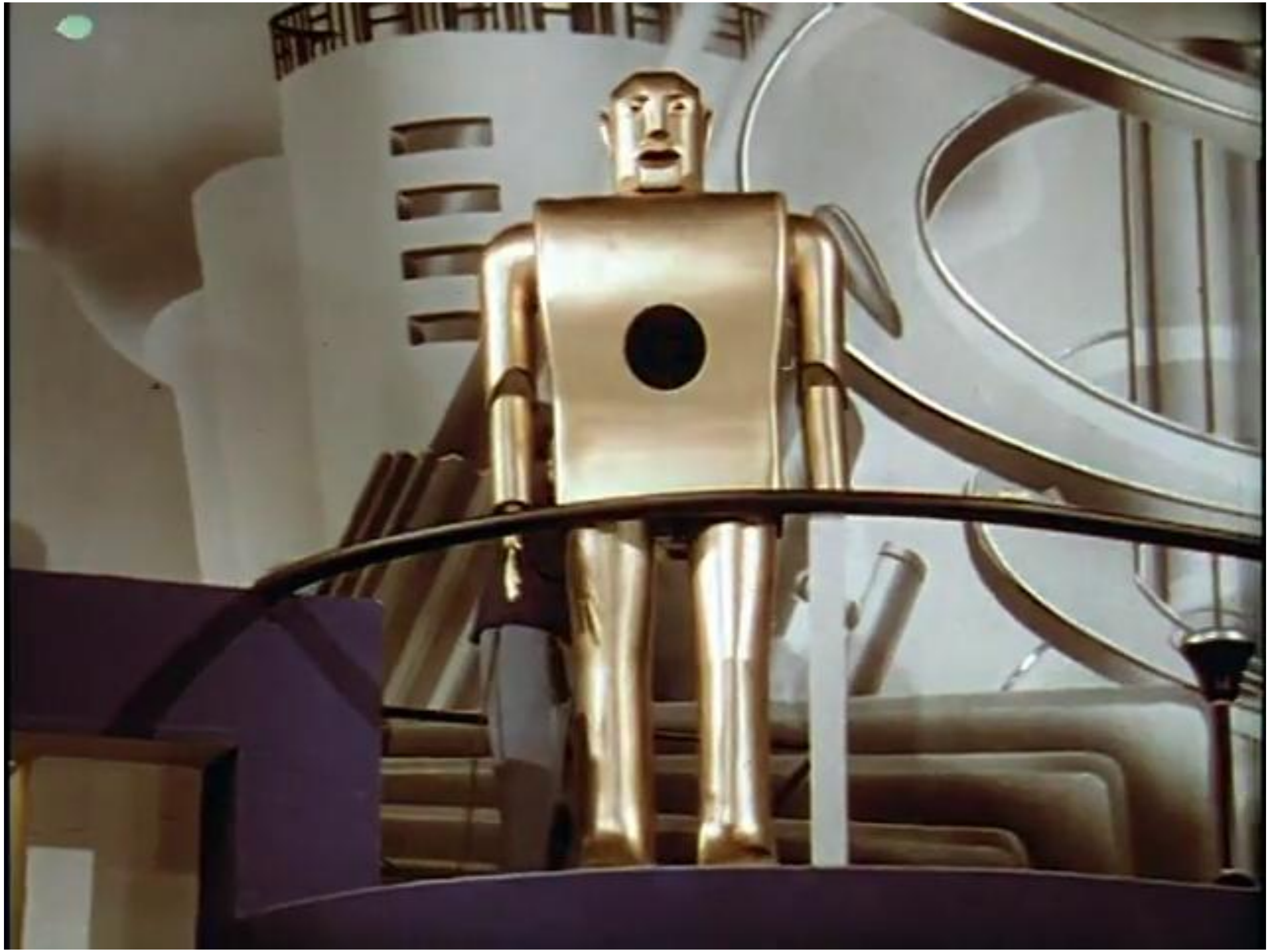
1939 – 1940

- Westinghouse presents the humanoid robot Elektro at the 1939 New York World's fair
- Elektro can speak, walk, move its head, and recognize colors through a photocell
- 26 pre-programmed actions can be triggered by speech
- In 1940, Westinghouse develops Sparko, a robot dog that is commanded by Elektro
- To many spectators, Elektro appears like a living creature and becomes a huge success in popular culture for two decades

Image courtesy of Computer History Museum. © Westinghouse Electric Corporation.

Elektro in Action

"I have a very fine brain of 48 electrical relays"



Scene from the movie

"The Middleton Family at the New York World's Fair"

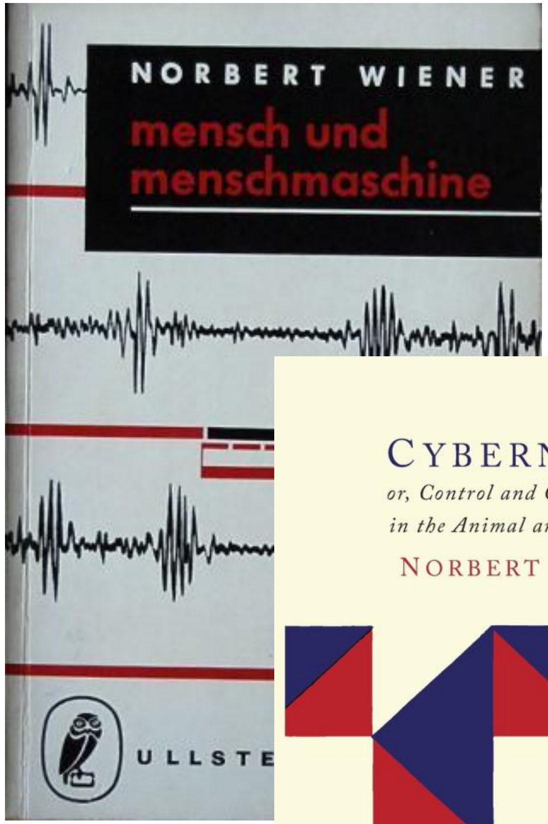
Westinghouse Electric & Manufacturing Co., 1939

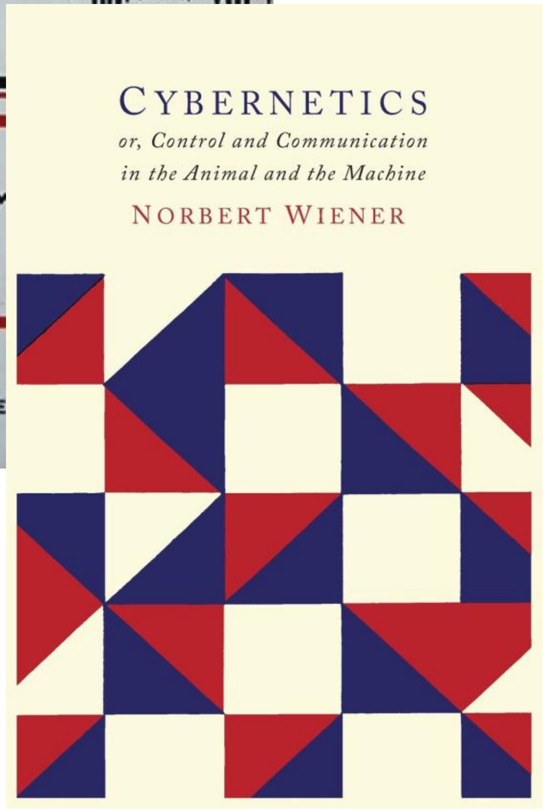
Public Domain

Robotics and Neuroscience

The First Autonomous Robot was Built to Study Brain-Like Cognition

- In the early years of robotics, computers were still in their infancy
- The first robots were mechanical automata with simplistic behavioral patterns
- The foundation of Cybernetics through Norbert Wiener in 1948 laid the theoretical foundations for "the scientific study of control and communication in the animal and the machine"
  (Wiener, 1948)
- The first autonomous robot was constructed by neurophysiologist William Grey Walter to study the emergence of complex behavior in a simple control system based on only two radio tubes





Images from https://library.bfg-muenchen.de/node/96 and Amazon

Bioinspired Robots with “Brains”

1950

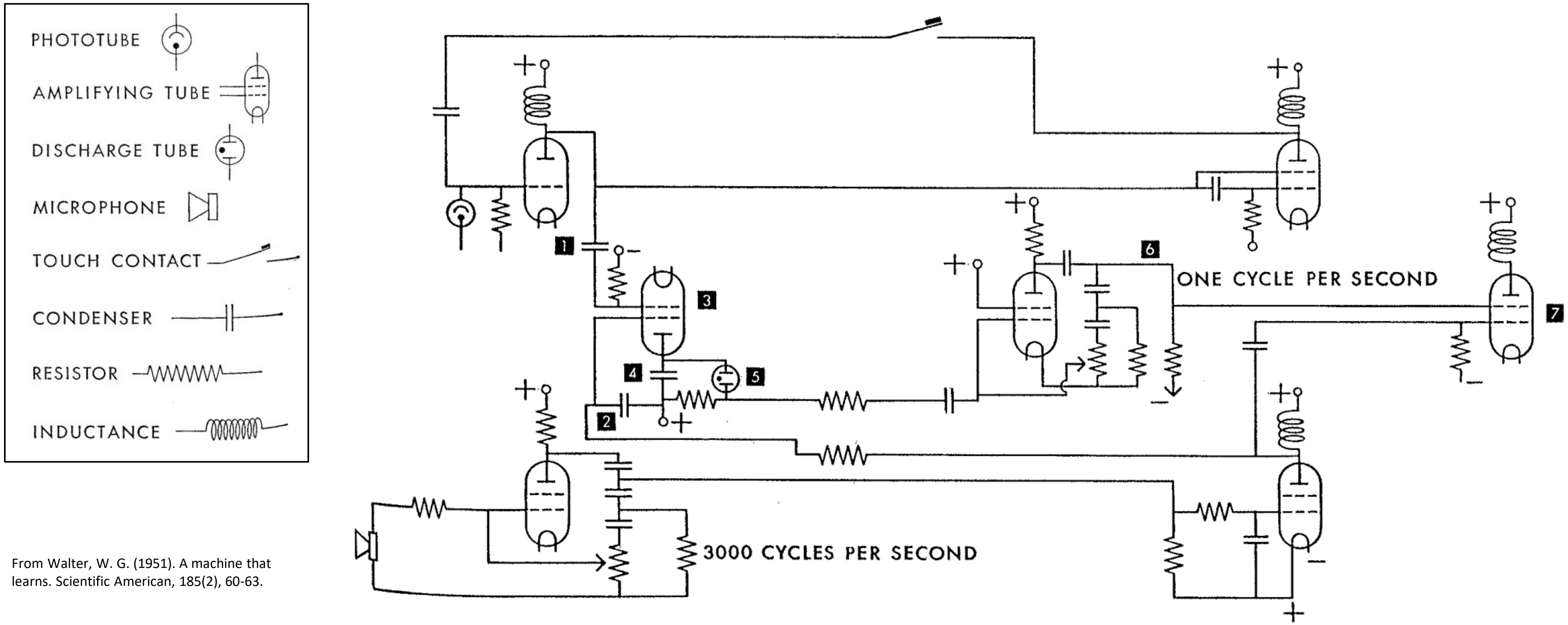
In his work “An Imitation of Life”, neurophysiologist William Grey Walter presents Elmer and Elsie, two machines of his artificial “robot species” *Machina speculatrix*. The robots implement phototaxis and automatically return to their charging station before batteries are empty.

- “Sense organs” for light and touch
- A nervous system simulated by two radio tubes
- Two motors for crawling and steering
- The direction of the photo sensor is controlled by the steering motor

Elmer and Elsie in Action

From https://www.youtube.com/watch?v=lLULRlmXkKo

The “Brain” of William Gray Walters Tortoises



From Walter, W. G. (1951). A machine that learns. Scientific American, 185(2), 60-63.

The “brain” of the robots developed by William Gray Walter is based on an analog electrical circuit (CORA, conditioned analog reflex) that emulates conditioned reflexes.

Bioinspired Robots with “Brains”

1951

Edmund C. Berkeley develops Squee, a robotic squirrel that collects nuts and carries them to a nest.

- “Sense organs” (sensors)
  - Two phototubes
  - Two contact switches
- “Acting organs” (motors)
- “Small brain” (relays)

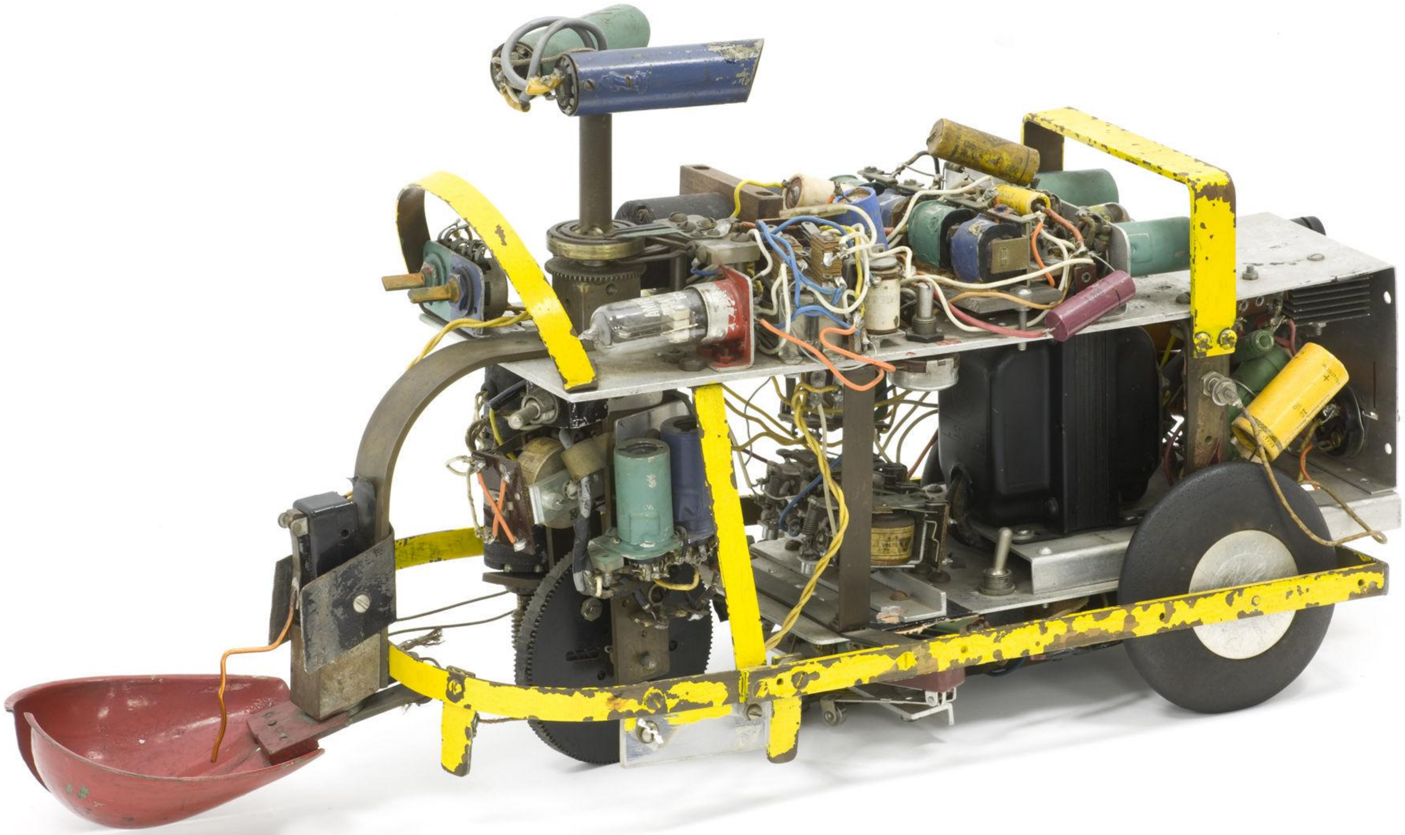


Image courtesy of Computer History Museum and Gordon and Gwen Bell. © Mark Richards

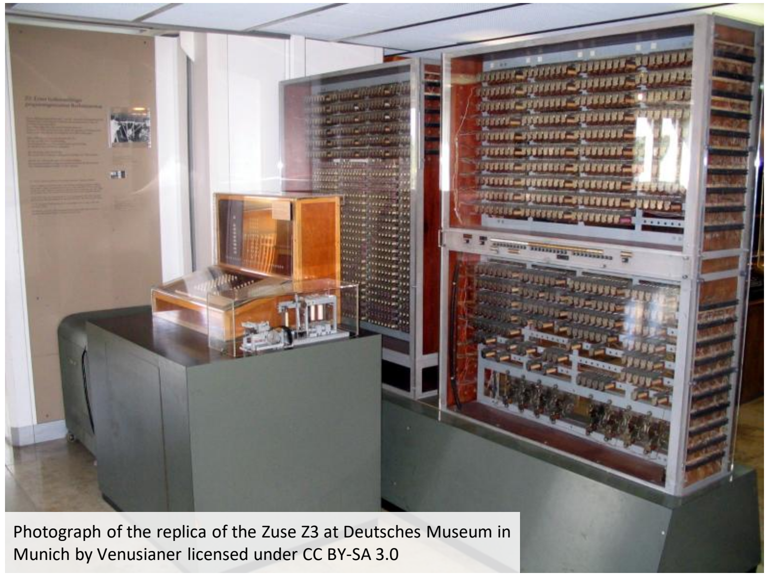
More Power for Artificial Brains: The Advent of Digital Computers

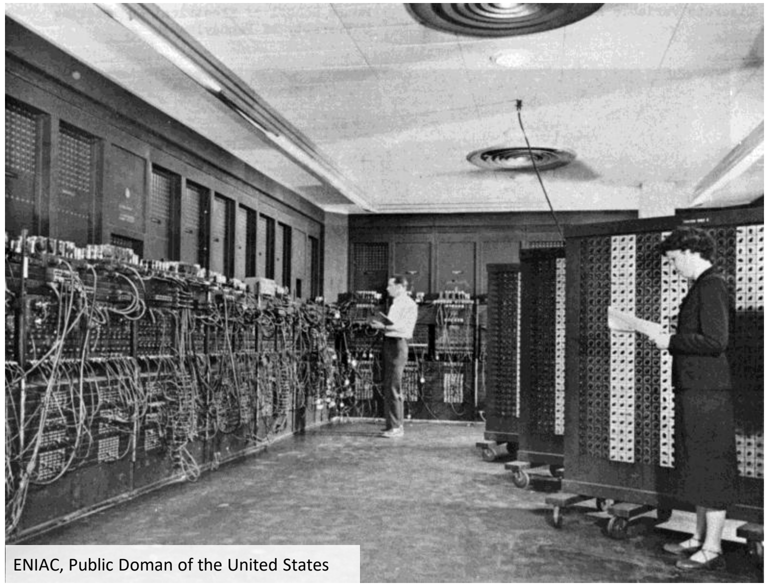
1837
Charles Babbage designs the Analytical Engine, the first Turing-complete computer

1935 – 1938
The computers Z1, Z2 and Z3 constructed by Konrad Zuse support floating point arithmetic

1946
ENIAC is a Turing-complete programmable electronic computer with 18,000 vacuum tubes

1971
Release of the Intel 4004, the first commercially available microprocessor


Photograph of the replica of the Zuse Z3 at Deutsches Museum in Munich by Venusianer licensed under CC BY-SA 3.0


ENIAC, Public Doman of the United States

Computers – Machines that Think?

In his book "Giant Brains or Machines that Think" from 1949, Edmund C. Berkeley describes computers as machines that are similar "to what a brain would be if it were made of hardware and wire instead of flesh and nerves"

- The first computers automated and accelerated tedious calculations that used to be done manually or that were not infeasible before
- It took many years to develop the algorithms, tools and interfaces that enabled computers to become the "brains" of robots
- Together with computer vision, artificial intelligence, and machine learning, robotics has become its own engineering discipline with hardly any remaining links to neuroscience

1956: The Birth of AI

John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon

A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug. 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

- **Automatic Computers**
- **Natural Language Processing**
- **Neural Networks**
- **Theoretical considerations about the scope of computer operations**
- **Self-Improvement**
- **Abstractions**
- **Randomness and Creativity**

The Advent of Modern Machine Learning

1955

Arthur Samuel implements a computer program for playing checkers that is bases on machine learning.

The system is capable of improving by playing against itself and wins against a human champion in 1962.

Some Studies in Machine Learning Using the Game of Checkers

Arthur L. Samuel

Abstract: Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these experiments are, of course, applicable to many other situations.

Introduction

The studies reported here have been concerned with the programming of a digital computer to behave in a way which, if done by human beings or animals, would be described as involving the process of learning. While this is not the place to dwell on the importance of machine-learning procedures, or to discourse on the philosophical aspects,[1] there is obviously a very large amount of work, now done by people, which is quite trivial in its demands on the intellect but does, nevertheless, involve some learning. We have at our command computers with adequate data-handling ability and with sufficient computational speed to make use of machine-learning techniques, but our knowledge of the basic principles of these

method should lead to the development of general-purpose learning machines. A comparison between the size of the switching nets that can be reasonably constructed or simulated at the present time and the size of the neural nets used by animals, suggests that we have a long way to go before we obtain practical devices.[2] The second procedure requires reprogramming for each new application, but it is capable of realization at the present time. The experiments to be described here were based on this second approach.

- *Choice of problem*

For some years the writer has devoted his spare time to

1956: Playing Checkers with AI on the IBM 701

"On February 24, 1956, Arthur Samuel's Checkers program, which was developed for play on the IBM 701, was demonstrated to the public on television. In 1962, self-proclaimed checkers master Robert Nealey played the game on an IBM 7094 computer. The computer won. Other games resulted in losses for the Samuel Checkers program, but it is still considered a milestone for artificial intelligence, and offered the public in the early 1960s an example of the capabilities of an electronic computer."

https://www.ibm.com/ibm/history/ibm100/us/en/icons/ibm700series/impacts/

IBM 701
Electrostatic storage with 2048 words of 36 bits; two programmer accessible registers; base cycle time of 12 µs

"It is not my aim to surprise or to shock you – but the simplest way I can summarize is to say that there are now machines in the world that think, that learn, that can create." (1957)

Herbert A. Simon (1916 – 2001)

AI – A new Field of Research?

Artificial Intelligence draws from many other disciplines such as mathematics, logic, control theory, decision theory etc. What sets it apart from these fields is …

- … the goal of duplicating human faculties (language, learning, memory, rational thinking, creativity etc.).
- … the application of computer science to implement these faculties as working computer programs for machine that operate in real-world environments.

Problem Solvers – Early Applications of AI

The first AI systems such as the Logical Theorist were designed to solve general problems that were defined in formal languages such as logic. They targeted tasks with very narrow scopes and became known as microworlds. Examples include:

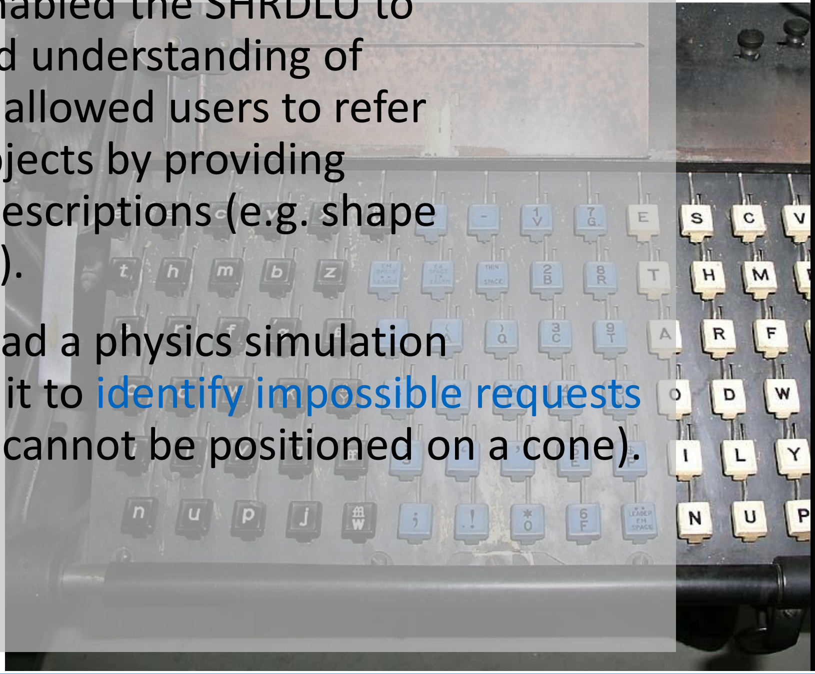
- Mathematical proofs
- Strategic games (chess, checkers)
- Simple geometric problems (“blocks world”)

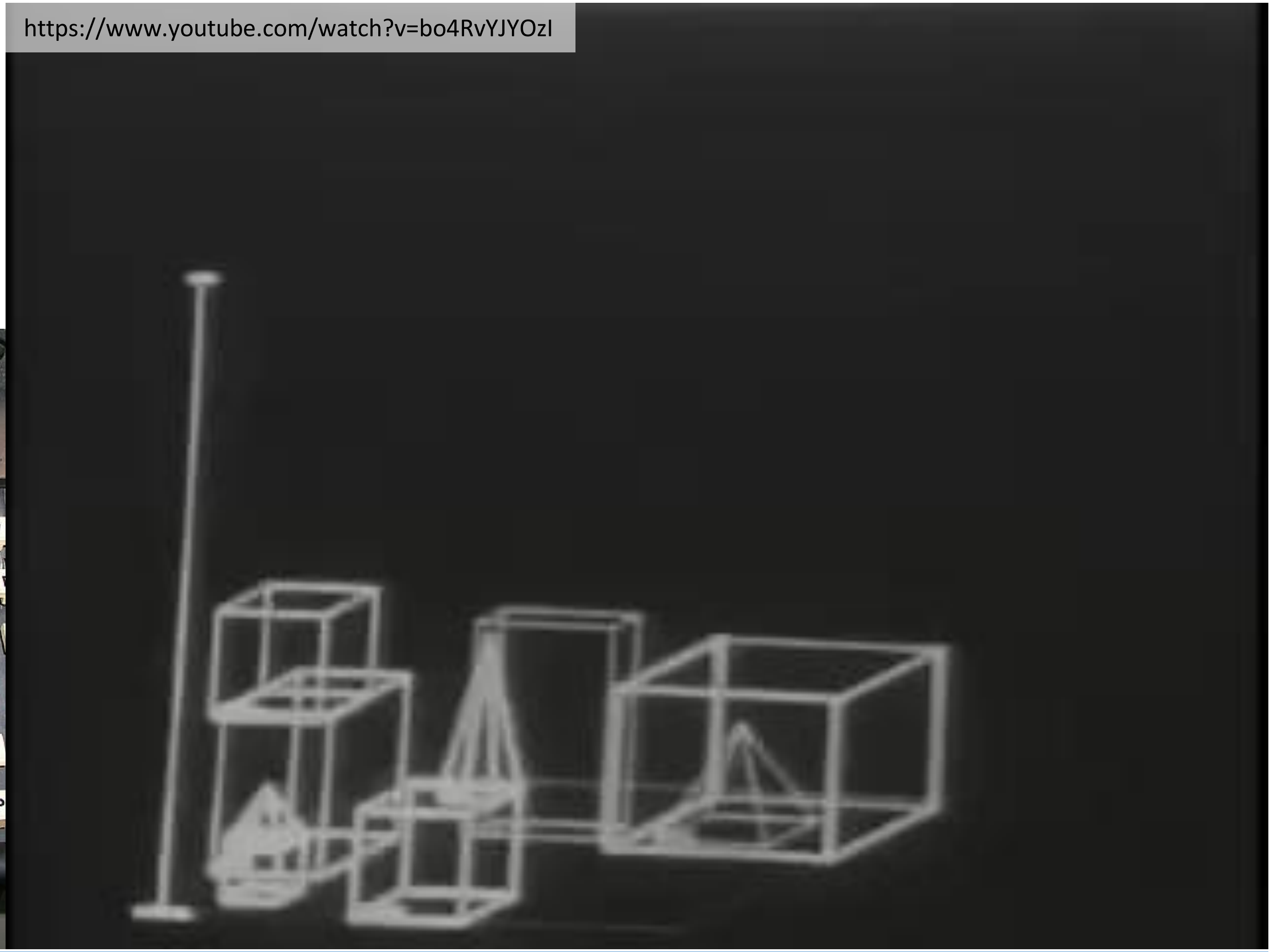
Example: Geometric Queries in Blocks World

1968 - 1973
The AI system SHRDLU allowed the interaction with world comprised of simple geometric objects through natural language (e.g. move objects from one position to another).

A memory enabled the SHRDLU to have a limited understanding of context. This allowed users to refer to specific objects by providing incomplete descriptions (e.g. shape but not color).

The system had a physics simulation that enabled it to identify impossible requests (e.g. a block cannot be positioned on a cone).





Towards Intelligent Robots

Courtesy SRI International, © SRI International

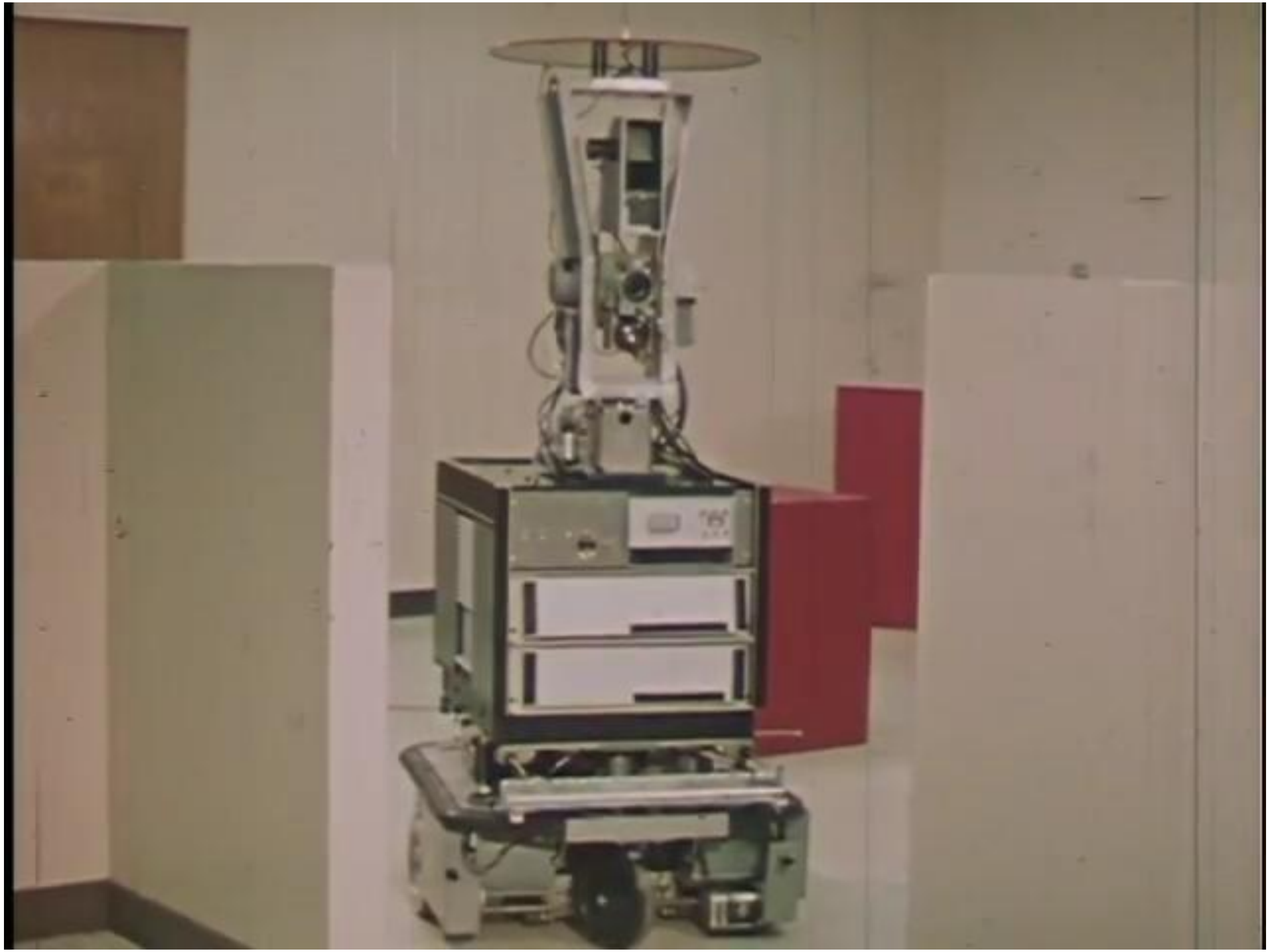
1961
The first industrial robot is installed at a production plant of General Motor.

1966 – 1972
The mobile robot Shakey is developed at SRI for the first time combined computer vision, language processing and the planning system STRIPS to navigate in a simplified world with blocks and ramps.

Shakey in Action

Video and text from https://youtu.be/GmU7SimFkpU

“From 1966 through 1972, the Artificial Intelligence Center at SRI International (then Stanford Research Institute) conducted research on a mobile robot system nicknamed “Shakey.” Endowed with a limited ability to perceive and model its environment, Shakey could perform tasks that required planning, route-finding, and the rearranging of simple objects.”



1969: Expert Systems

- General problem solving systems do not have any specific knowledge about the application domain (“weak methods”)
- Solving more complex problems beyond microworlds turned out to be infeasible due to large requirements for compute power
- Expert systems include domain-specific knowledge for the targeted application, which considerably simplifies inference processes
- Early examples include the inference of molecular structures from mass spectrograms, medical diagnosis systems or configurations of orders for computer systems

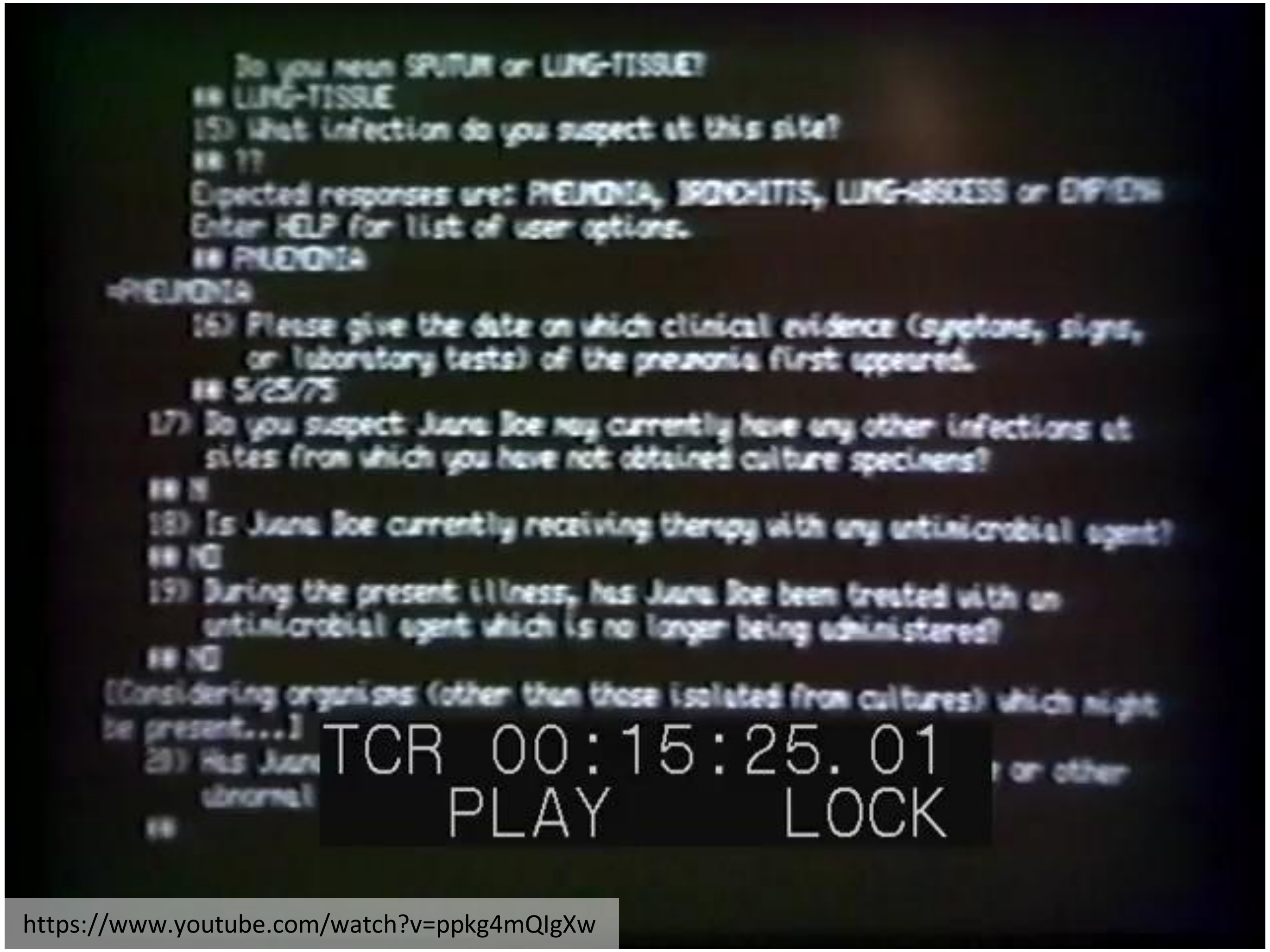
Example: Mycin (1975)

Mycin was an expert system for identifying infectious bacteria and recommending appropriate antibiotics together with the patient-dependent dosage. The system …

… was based on around 600 rules

… provided a sequence of textual queries to the physician

… incorporated a notion of uncertainty through so-called certainty factors



https://www.youtube.com/watch?v=ppkg4mQIgXw

Example: OPS5, R1/XCON (1980)

OPS5 is a rule-based programming language that was used to implement the R1/XCON expert system that was capable of automatically configuring the VAX computer systems by DEC based on customer requirements. The system …

… initially deployed in 1980

… contained about 2500 rules

… had processed 80,000 orders by 1986 with up to 95 % accuracy

… and saved estimated 25 million USD per year

Example: DEC VAX-11/780

https://www.computerhistory.org/revolution/mainframe-computers/7/182/736

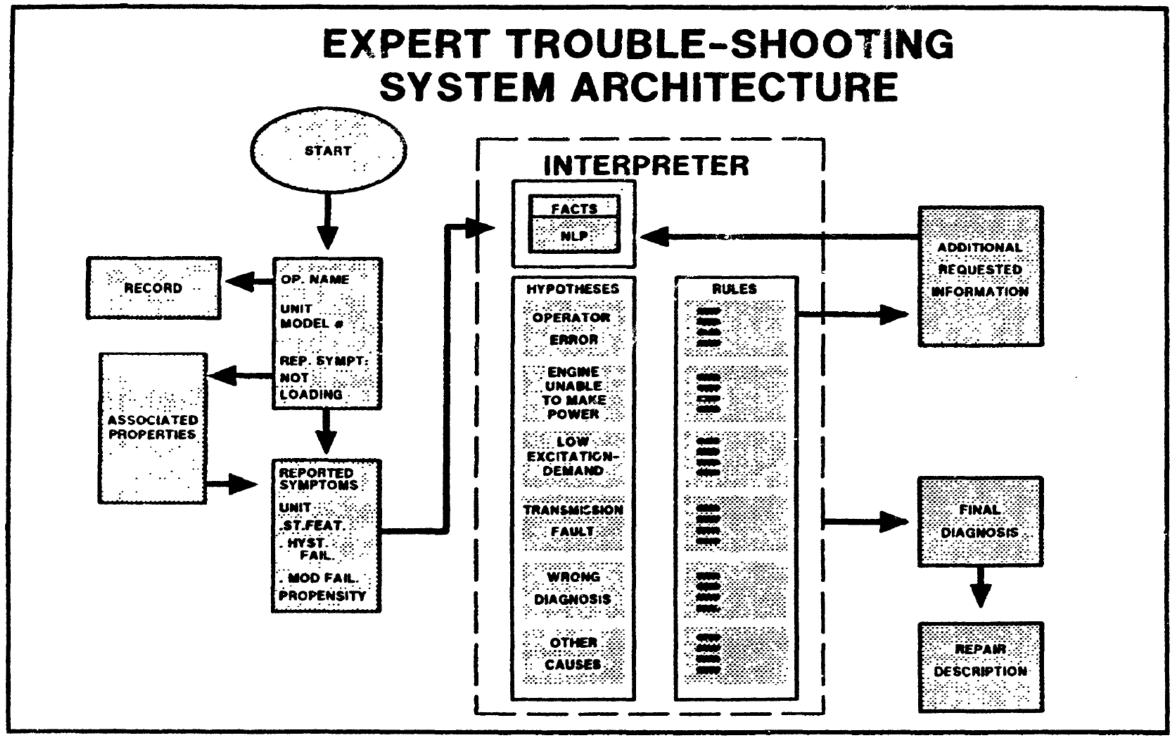
Example: Delta (1983)

Delta was an expert system for troubleshooting and repairing diesel electric locomotives. The system was implemented on a PDP 11/23 and

… contained about 530 rules

… represented the knowledge of a senior field engineer

… provided textual information, CAD diagrams and repair sequences from a video disk as support



Example: DEX.C3 (1985)

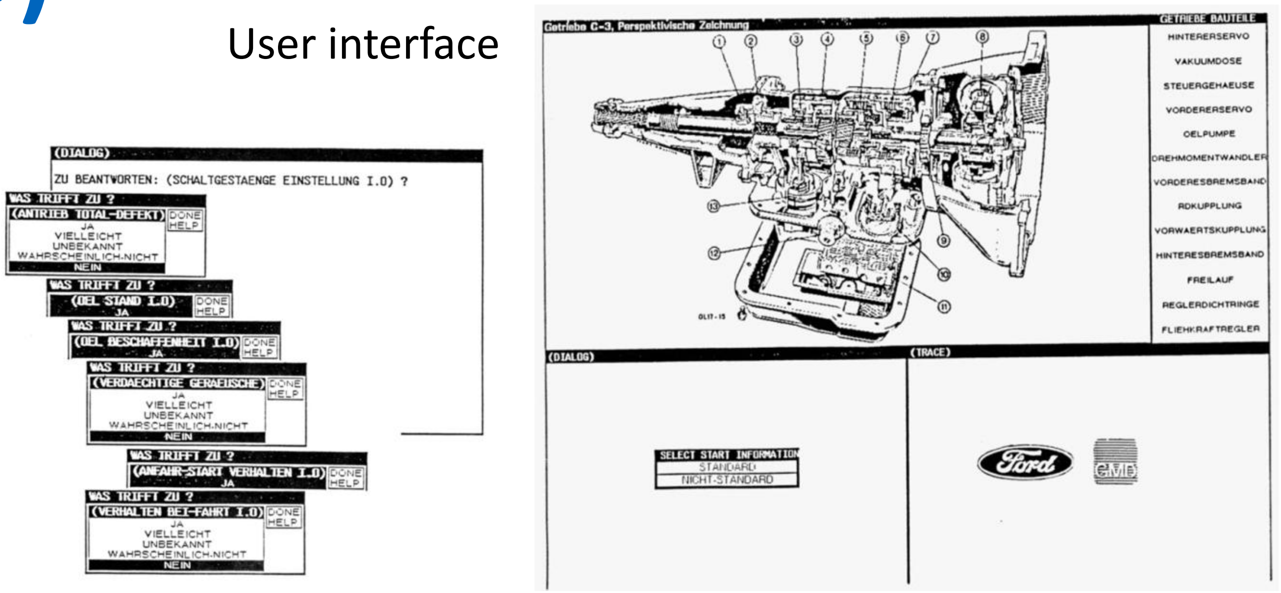
DEX.C3 was an expert system for error diagnosis in the C3 gearbox from Ford. It was implemented in Germany on a XEROX 1108 computer. The system …

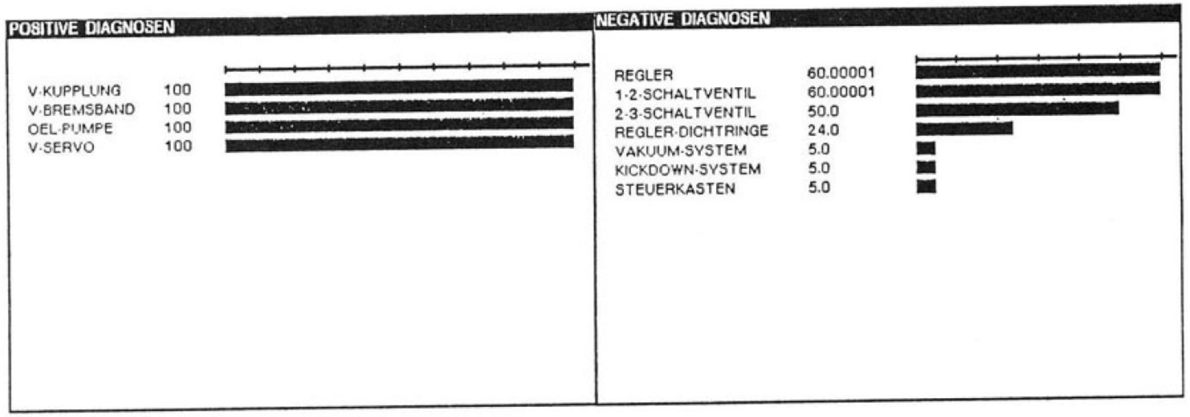
… contained 140 rules

… used certainty factors

… was able to explain to the user why it made a certain decision

User interface





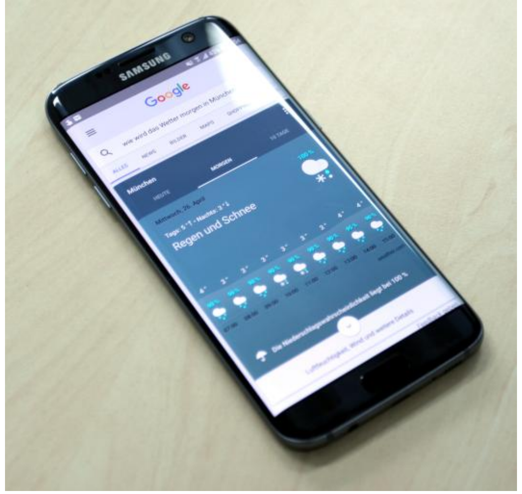
V-KUPPLUNG	100
V-BREMSBAND	100
OEL-PUMPE	100
V-SERVO	100

REGLER	60.00001
1-2-SCHALTVENTIL	60.00001
2-3-SCHALTVENTIL	50.0
REGLER-DICHTRINGE	24.0
VAKUUM-SYSTEM	5.0
KICKDOWN-SYSTEM	5.0
STEUERKASTEN	5.0

Final diagnosis output produced by the system.

Cognition – A Solved Problem?

- Industrial robots surpass human workers in precision, speed and reliability
- Computers defeat human competitors in challenging games
  - In 2011, IBM's cognitive computing platform Watson wins in the U.S. quiz show Jeopardy against two human competitors
  - In 2016, DeepMind's AlphaGo beats Go champion Lee Sedol
- Speech recognition, text translation, image recognition, and many other tasks that were formerly done by humans are now available on every smartphone

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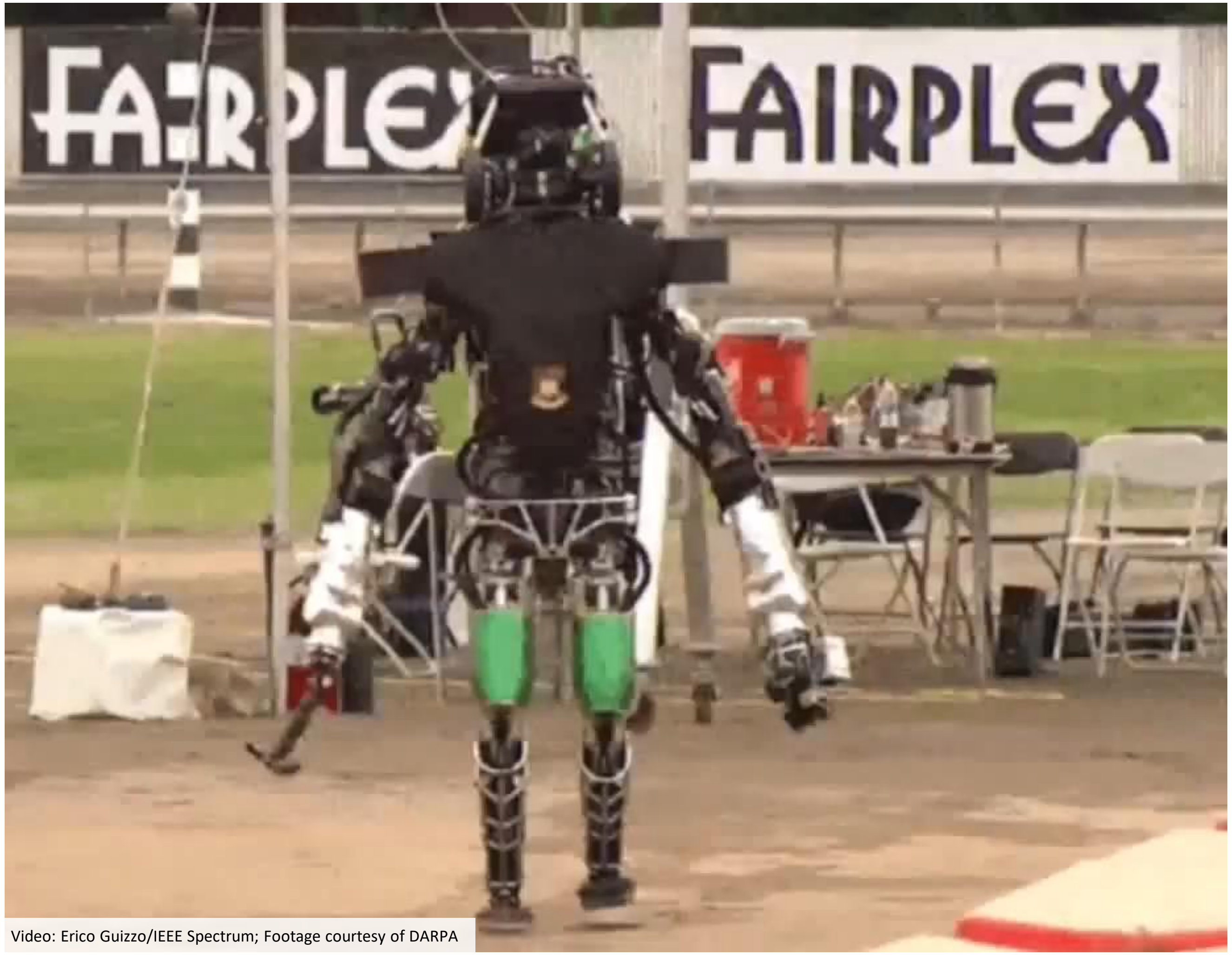
Photograph by Vanillase licensed under CC BY-SA 3.0

Cognition – An Unsolved Problem!

- All cognitive systems built so far are highly specialized only for specific niches
- Living creatures are more versatile and flexible than any robot ever built
- There is no unified definition or theory of what cognition is and what it means
- The design of cognitive systems is both the application and at the same time also the extension of what is known about cognition



Video: Erico Guizzo/IEEE Spectrum; Footage courtesy of DARPA

Are Cognitive Systems a Threat?

“The development of full artificial intelligence could spell the end of the human race. We cannot quite know what will happen if a machine exceeds our own intelligence, so we can't know if we'll be infinitely helped by it, or ignored by it and sidelined, or conceivably destroyed by it.”

Stephen Hawking, 2014 (BBC)

“I think we should be very careful about artificial intelligence. If I had to guess at what our biggest existential threat is, it’s probably that. So we need to be very careful.”

Elon Musk, 2014 (The Guardian)

Cognitive Systems are Tools!

artificial intelligence is a tool, not a threat

“Recently there has been a spate of articles in the mainstream press, and a spate of high profile people who are in tech but not AI, speculating about the dangers of malevolent AI being developed, and how we should be worried about that possibility. I say relax. Chill. This all comes from some fundamental misunderstandings of the nature of the undeniable progress that is being made in AI, and from a misunderstanding of how far we really are from having volitional or intentional artificially intelligent beings, whether they be deeply benevolent or malevolent. […]”

Rodney Brooks, 2014 (rethink robotics)

Summary

- Humankind has always strived to build machines with cognitive capabilities similar to those of living creatures
- The first autonomous robots were built by a neurophysiologist in order to study the function of the brain
- Early computers were described as "mechanical brains"
- Over time, computer science and robotics have evolved to independent fields of research with an its engineering-based view of cognition that is very different from that applied in early brain-inspired systems
- Today, different theories, models, and artificial implementations of cognition compete with each other